**MATH 195: 11/10**
**SOLVING TRIG EQUATIONS**

**The basic case.**

To solve equations involving trig functions, we need to know how to solve equations of the form $\mathsf{trig}(t) = A$. For these the unit circle picture is the guide.

- For each, one solution $t_0$ is given by $\mathrm{arctrig}(A)$.
  - For cos this solution is on the top half of the circle.
  - For sin and tan this solution is on the right half of the circle.
- A second solution in another quadrant is found by a geometric transformatioon.
  - For cos, do a vertical reflection to get the bottom half solution. In symbols, this is $-t_0$.
  - For sin, do a horizontal reflection to get the left half solution. In symbols, the this is $\pi - t_0$.
  - For tan, rotate by a half circle to get the left half solution. In symbols, this is $\pi + t_0$.
- The general solution is found by adding integer multiples of $2\pi$ to these two solutions: $+2\pi k$, where $k \in \mathbb{Z}$.
  - For tan you can combine these last two steps. The general solution is integer multiples of $\pi$ added to the first solution $t_0$: namely, $t_0 + \pi k$, where $k \in \mathbb{Z}$.

Pay attention to the range of trig functions; $\mathsf{trig}(t) = A$ may have zero solutions.

**Solving trig equations of $\mathsf{trig}(t)$.**

Your intermediate goal is to get turn the equation into the form $\mathsf{trig}(t) = A$. Then you solve it by the above methods. If you have an equation with the secondary trig functions, rewrite in terms of the primary trig functions.

EXERCISES, PART 1

(1) Find the general solution of $\cos t = 1/2$.
(2) Find the general solution of $\sin t = 0$.
(3) Find the general solution of $\tan t = 2$.
(4) Find the general solution of $3\cos x = 2$.
(5) Find the general solution of $2\csc y + 4 = 0$.
(6) Find the general solution of $\sin u = \cos u$. [Hint: you can rewrite to be about a single trig function.]
(7) Find the general solution of $2\cos^2 t - \cos t = 0$, by following this outline of steps.
   - This is a quadratic equation in the "variable" $\cos t$. As a first step, use factoring or another method to solve the equation for $\cos t$. As this is a quadratic equation, you should expect to see up to two solutions.
   - Now you have one or more equations of the form $\cos t = A$ to solve. Solve them.
   - Combine together the general solutions for each of these to get the general solution the original equation.
(8) Find the general solution of $\sin^2 x + \sin x - 2 = 0$ by following a similar process.
(9) Find the general solution of $\tan(x)\sin(x) = 0$.

**Solving trig equations with complicated inputs.**

If the input to trig isn't just a variable, you have an extra step at the end.

- First get the equation in the form $\mathsf{trig}(E(t)) = A$, where $E(t)$ is the input in terms of the variable $t$ you are solving for.
- Solve to find the general solution for $E(t)$ using the previous method.
- Once you know $E(t)$, solve for $t$.
  - For example, if $E(t)$ is of the form $mt + b$, you need to add $b$ to both sides then divide both sides by $m$. When you divide, be sure to divide the whole equation, including the $+2\pi k$ part.
  - If $E(t)$ is more complicated than a linear expression, you will have to use other methods, as appropriate.

## Exercises, part 2

(1) Find the general solution of $\cos(2t) = 1/3$ by first solving for $2t$, then using that to solve for $t$.
(2) Find the general solution of $2\sin(3t) + 1 = 0$.
(3) Find the general solution of $3\tan(t + \pi/2) = 1$.
(4) Find the general solution of $\sin^2(t/2) = \sin(t/2)$. [Hint: this is a quadratic equation in $\sin(t/2)$, so you should expect to have two equations of the form $\sin(t/2) = A$ to solve.]
(5) Find the general solution of $\cos(\pi t - \pi/4) = 1/3$.
(6) Find the general solution of $2\tan(t/2 + \pi/8) - 1 = 0$.
(7) Find the general solution of $e^{\cos t} = 1$.
(8) Find the general solution of $\cos(e^t) = 1$.
(9) Find the general solution of $\tan(t^2) = 2/5$.
(10) Find the general solution of $\sin(\cos t) = 0$. [Hint: you will get infinitely many solutions for $\cos t$, but all except one of them will be out of range.]
(11) Start finding the general solution of $\sin(\tan t) = 1$ until you get to a point where it's too awful to continue. Write a sentence or two explaining what you would need to do to finish.
(12) Find all $x$-intercepts of the function
$$f(x) = 3\tan(2x).$$
(13) The depth in meters of an ocean wave over time in seconds is given by the formula
$$w(t) = 3\sin(\pi t/4) + 3.$$
Find all times $t \geq 0$ when the wave has a depth of 1.5 meters.